Comptia N+

N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

DAY5

TASKS:1

Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

RESULT

1. Opened Command Prompt.
2. Typed the following command: ping www.youtube.com
3. Pressed Enter.
4. Waited for the ping process to complete.
5. Noted the average value for your own computer.

TASKS:2

Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Each hop represents a network device your data passes through to reach the destination.

RESULT

1. Opened Command Prompt.
2. Typed: tracert www.flipkart.com
3. Pressed Enter.
4. Wait for command to finsh
5. Note down each hop ip address or hostname

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TASKS:3

Use the nslookup tool to find the IP address of www.zomato.com and record the result.

RESULT

1. Opened Command Prompt.
2. Typed: nslookup www.zomato.com
3. Press Enter
4. Recorded the IP address return by the DNS server

TASKS:4

Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

RESULT

1. Connected my laptop and phone to the same Wi-Fi network.
2. Found the phone's IP address.
3. Opened Command Prompt.
4. Typed: ping 192.168.x.x
5. Pressed Enter.
6. Observed the results.
7. Turned off the phone's Wi-Fi and repeated the ping test.

OBSERVATION:

When the device was online, the ping command received replies successfully.

When the device was offline, the ping command displayed messages such as Request timed out.

OR

Destination host unreachable.

TASKS:5

Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems:

(a) Website not loading

(b) Slow connection

(c) Suspecting DNS issues.

Give a specific example for each tool based on your own test results above.

RESULT

1. **Ping** is most useful for checking whether a website or network host is reachable and measuring response time.
2. For example, ping www.youtube.com can show whether YouTube responds and the average response time.

3. **Traceroute (tracert)** is useful for finding where a connection becomes slow or fails along the network path.
4. For example, tracert www.flipkart.com shows the intermediate hops between my computer and Flipkart.
5. **Nslookup** is most useful for diagnosing DNS issues by checking whether a domain name resolves to an IP address.
6. For example, nslookup www.zomato.com displays the IP address returned by the DNS server.